



Explorative Data Analysis - Genk

Leveraging Data Analytics for Efficient OR Planning at ZOL

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Contents

Introduction	3
Descriptive analysis	3
Temporal information	3
Year	3
Weekday	4
StartHour	4
Clinical characteristics	5
ProcedureName	5
AdmissionType	5
UrgencyType	6
Hospital stay	6
LengthOfStay	6
Planned surgery information	8
PlannedDurationMinutes	8
PlannedOR	9
Actual surgery information	10
ActualOR	10
DurationMinutes	12
Deviation metrics	13
StartDiff	13
EndDiff	15
percentage_deviation	19
Conclusion	22

Introduction

This document provides an exploratory overview of the dataset used for analyzing operating room (OR) efficiency at Ziekenhuis Oost-Limburg, campus Genk. The aim is to describe the main temporal, clinical, and operational characteristics of the data before moving toward more specific performance metrics. By systematically reviewing the distributions, frequencies, and summary statistics of key variables, we gain a first impression of how surgical activity is structured across time, departments, and procedure types.

The analysis follows a descriptive approach, highlighting patterns that may explain later deviations between planned and observed activity. Each visualization or table is accompanied by a short interpretation that connects numerical trends to the daily practice of the operating theatre. Together, these insights form the empirical foundation for subsequent modelling and the development of operational key performance indicators (KPIs) focused on scheduling accuracy, resource allocation, and overtime work.

Descriptive analysis

The descriptive analysis provides a structured overview of the main variables of the dataset before moving to more advanced comparisons between planning and execution. It summarizes how surgical activity is distributed across time, procedure types, and operational variables, offering a factual foundation for interpreting later results. By examining these baseline distributions, we can better understand the context in which deviations, overtime, and scheduling inefficiencies occur.

Temporal information

Understanding when surgeries take place is essential for interpreting overall operating room performance. Temporal variables provide insight into how surgical activity is distributed across years, weekdays, and hours of the day. These patterns reflect both organizational planning and patient demand, and they influence key operational factors such as staff allocation, overtime work, and resource utilization. The following subsections explore these temporal dimensions to establish a baseline view of how surgical activity is structured over time.

Year

Surgical activity remained relatively stable over the observed period, with a slight increase from 27.9 percent in 2022 to 29.9 percent in 2024. The distribution across these years is well balanced, indicating a consistent surgical volume over time. The lower proportion observed in 2025, accounting for 12.5 percent of all procedures, is expected, as the data only cover a partial year up to the moment of extraction. Table 1 provides the yearly counts and percentages.

Table 1: Number of surgeries per year (count and percentage)

Year	n	Percentage
2022	22133	27.9%
2023	23575	29.7%
2024	23738	29.9%
2025	9906	12.5%

Weekday

Surgical activity is clearly concentrated during regular weekdays, with each weekday from Monday to Friday accounting for approximately one fifth of all procedures. The distribution is relatively balanced, ranging from 17.9 percent on Mondays to 20.0 percent on Thursdays. Weekend activity remains limited, with 2.1 percent of procedures performed on Saturdays and 1.8 percent on Sundays. This pattern aligns with the hospital’s scheduling policy, where weekends are primarily reserved for urgent or emergency cases. Table 2 provides the detailed counts and percentages.

Table 2: Number of surgeries per weekday (count and percentage)

Weekday	Count	Percentage
Do	15844	20.0%
Wo	15814	19.9%
Vr	15615	19.7%
Di	14750	18.6%
Ma	14232	17.9%
Za	1654	2.1%
Zo	1443	1.8%

StartHour

Most surgeries start between 07:00 and 15:00, which corresponds to regular operating room hours. A pronounced peak is observed at 08:00, reflecting the start of the daytime operating schedule when the majority of planned procedures begin. Following this peak, the number of surgeries gradually declines throughout the late morning and afternoon. After 16:00, only a limited number of procedures are initiated, with very low activity during the evening and night. This distribution highlights a strong concentration of surgical activity within standard daytime hours, with after-hours activity remaining relatively scarce. Figure 1 presents the full distribution of start hours.

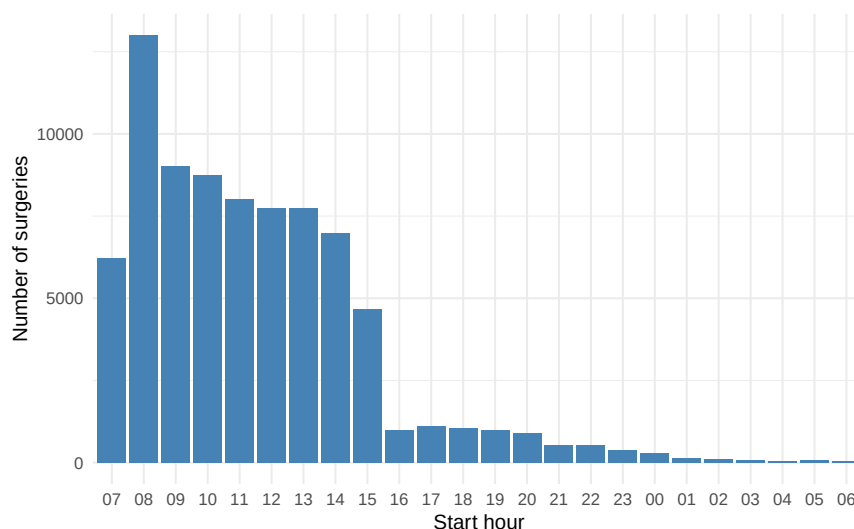


Figure 1: Distribution of surgery start times

Clinical characteristics

Clinical characteristics describe the type of surgical activity captured in the dataset. These variables outline what procedures were performed, under which admission conditions, and with what urgency level. Together, they provide an overview of the clinical composition of the surgical workload and highlight how case types and patient pathways influence planning complexity. Understanding this context is essential before interpreting operational performance or scheduling outcomes.

ProcedureName

The dataset includes a total of 1275 unique procedure names. On average, each procedure type was performed 62.1 times, while the median frequency equals 11 cases. This substantial difference between the mean and the median indicates a highly skewed distribution of surgical procedures. A relatively small number of procedure types are performed very frequently, whereas the majority occur only sporadically. Table 3 provides an overview of these summary statistics.

Table 3: Overview of surgical procedures

Unique procedures	Average cases per procedure	Median cases per procedure
1275	62.1	11

The twenty most frequent procedure names together represent a substantial share of all surgeries. The bar chart clearly illustrates a steep drop-off in frequency, with the most common procedures occurring several thousand times, followed by a gradual decline across the remaining top-ranked procedures. Despite this decrease, even the lower-ranked procedures within the top twenty still account for a considerable number of cases, indicating that surgical volume is concentrated in a relatively small group of recurring interventions. Figure 2 visualizes the distribution of these most frequently performed procedures.

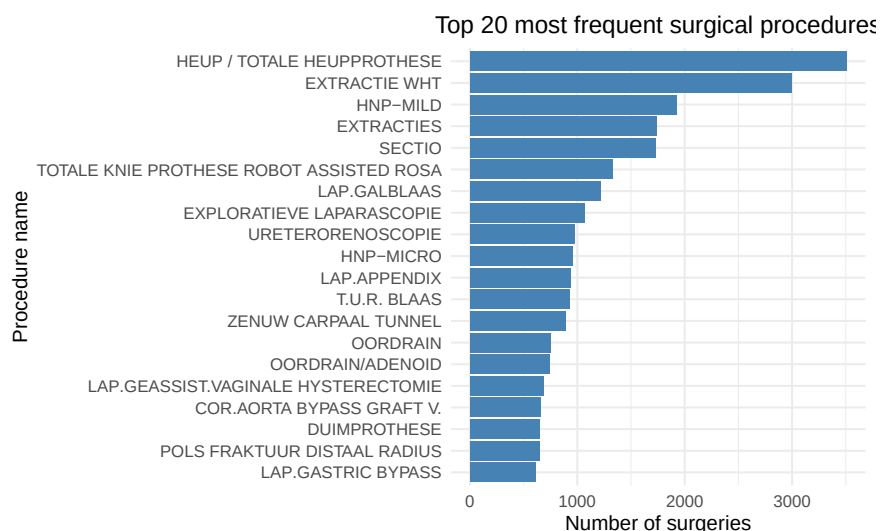


Figure 2: Top 20 most frequent surgical procedures

AdmissionType

Most surgeries in the dataset are registered as hospital admissions, representing 45798 procedures or 57.7 percent of all activity. Day admissions account for 33554 procedures, or 42.3 percent. This distribution

indicates that the majority of surgical activity takes place in an inpatient context, with a substantial share also performed as day cases. Table 4 provides the detailed counts and percentages.

Table 4: Number of surgeries per admission type (count and percentage)

AdmissionType	Count	Percentage
HOS	45798	57.7%
DAG	33554	42.3%

UrgencyType

The vast majority of surgeries are classified as elective, representing 67736 procedures or 85.4 percent of all cases. Non elective cases account for 11616 surgeries, or 14.6 percent. This distribution indicates that most procedures are scheduled in advance and follow a relatively predictable planning process. Non elective cases, although less frequent, can still have a significant operational impact since they often require last minute scheduling and may lead to deviations in the planned operating room program. Table 5 provides the detailed counts and percentages.

Table 5: Number of surgeries per urgency type (count and percentage)

UrgencyType	Count	Percentage
electief	67736	85.4%
niet-electief	11616	14.6%

Hospital stay

Hospital stay variables capture how long patients remain admitted after surgery. Analyzing these distributions helps identify whether short- or longstay cases dominate the dataset and how different admission or urgency types affect discharge timing. The results in this section provide a bridge between surgical scheduling and patient flow within the hospital.

LengthOfStay

The following analysis focuses only on hospitalized cases (AdmissionType = HOS). The average hospital stay after surgery is 7.65 days, with a median of 3 days, indicating that most patients are discharged quickly after their procedure. The distribution is highly skewed, as shown by the large difference between the mean and the median. While 95 percent of stays are shorter than 27 days, a small number of cases extend substantially longer, up to 301 days. These long stays represent rare outliers and likely reflect complex or rehabilitation related admissions rather than typical surgical recoveries. Table ?? summarizes the main summary statistics for length of stay.

```
## # A tibble: 1 x 10
##       n Mean Median   SD   IQR   Min   Max   P95   P99 P99_9
##   <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 45783  7.65     3 15.9     4     1   301    27  78.2  187
```

The distribution of hospital stay without revalidation confirms that most patients are discharged after a short admission. The frequency drops sharply after the lowest stay category, followed by a gradual decline

as length of stay increases. While short stays clearly dominate the dataset, a non-negligible number of cases remain hospitalized for several days, resulting in a right-skewed distribution with a long tail of extended stays. Figure 3 shows the truncated distribution.

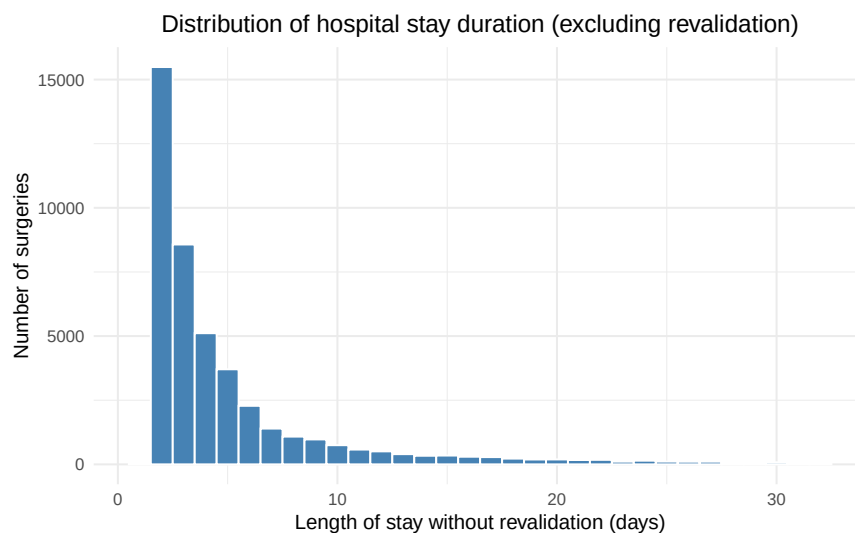


Figure 3: Distribution of hospital stay duration without revalidation

When comparing length of stay across urgency types, clear differences become visible. Non elective hospital admissions tend to have longer and more variable stays than elective ones, as reflected in the wider distributions and the more pronounced upper tail. Elective cases are more concentrated around shorter stays, whereas non elective cases show greater dispersion across longer durations. Overall, these patterns suggest that length of stay is influenced by the urgency of the surgery. Figure 4 provides an overview of these differences.

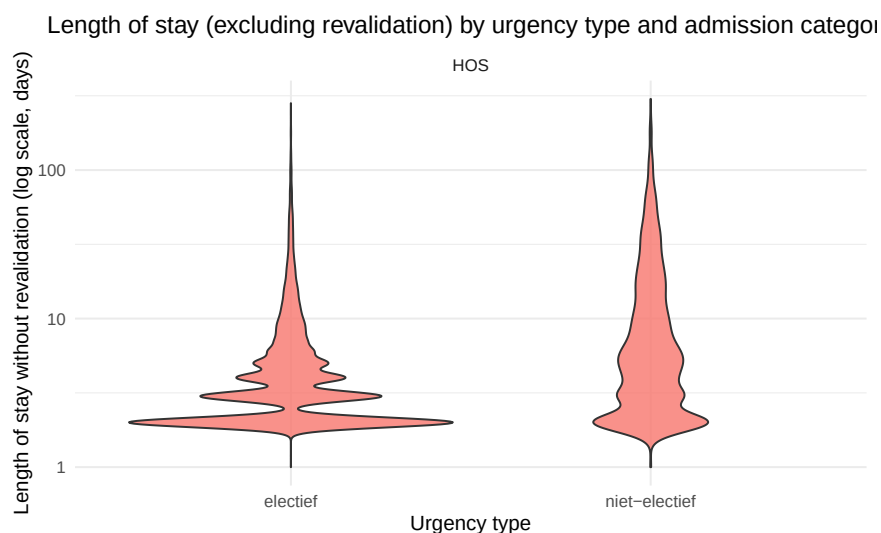


Figure 4: Length of stay without revalidation by urgency type and admission category

Planned surgery information

This section summarises all variables related to the planned part of the surgical workflow. It describes how surgeries were initially scheduled in terms of expected duration, operating room assignment, and distribution across rooms. Examining these metrics allows us to understand how operating capacity is allocated in advance and to what extent planning patterns differ between rooms. These insights serve as a reference point for comparing planned versus actual execution later in the report.

PlannedDurationMinutes

The planned duration of surgeries varies widely across the dataset. On average, procedures are scheduled for 94 minutes, while the median is 73 minutes, indicating that most surgeries are relatively short but a smaller group of long procedures increases the mean. The interquartile range of 68 minutes shows substantial variation within the middle 50 percent of cases. Most procedures are planned for relatively short to medium durations, although a small number extend to much longer planned times, with values up to 1420 minutes. At the upper end of the distribution, the ninety fifth percentile equals 250 minutes, the ninety ninth percentile 404 minutes, and the ninety nine point nine percentile 837.7 minutes. These rare outliers represent exceptionally long planned procedures. Table 6 summarises these statistics.

Table 6: Planned duration of surgeries in minutes, summary statistics

Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
94	73	84.63	68	4	1420	250	404	837.7

The distribution of planned surgery durations shows a strong right skewed pattern. Most procedures fall within the shorter duration categories, where the bars are clearly highest, after which the frequency gradually declines. Only a small fraction of surgeries appears in the longest duration ranges. This indicates that short and medium length procedures dominate the planned surgical workload, while long operations represent a relatively small portion of the overall schedule. Figure 5 shows the trimmed distribution up to the ninety ninth percentile.

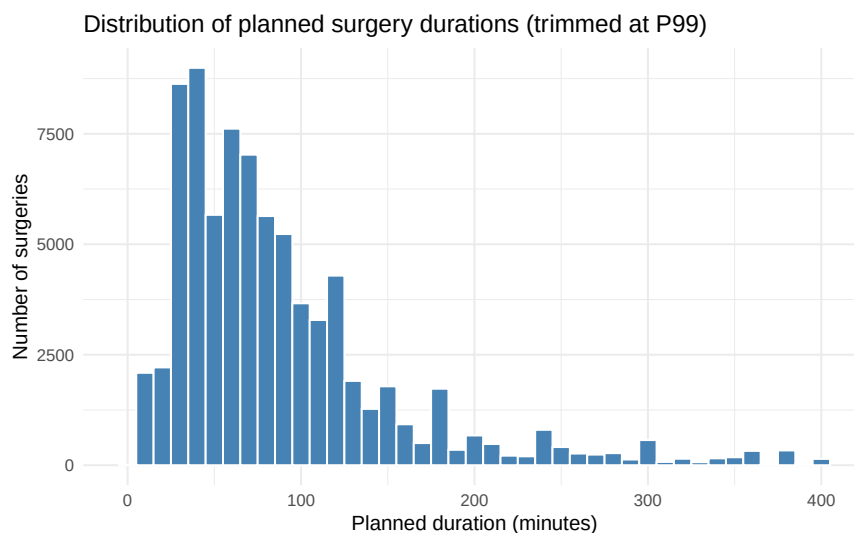


Figure 5: Distribution of planned surgery durations (trimmed at P99)

PlannedOR

The table below provides a detailed overview of surgical planning across all operating rooms. For each room, it lists the number of planned surgeries, summary statistics of planned durations, and the number of unique procedure names. This gives a comprehensive view of both workload and diversity in the planned surgical program. Table 7 summarises these metrics.

Planned activity is unevenly distributed across rooms. Some operating rooms handle several thousand planned surgeries, while others have only a very small number of cases. Average planned durations also vary considerably between rooms, with some rooms showing low mean values and others much higher ones. This variation indicates that certain rooms are used for shorter high throughput procedures, while others are scheduled for longer or more complex cases.

The coefficient of variation displayed in Table 7 shows substantial differences in variability between rooms. Procedure diversity likewise differs strongly, with some rooms registering only a few unique procedure names and others several hundred. Together, these patterns highlight that surgical planning at Ziekenhuis Oost Limburg is far from uniform across operating rooms, with clear differences in both workload and procedural variety.

Table 7: Per operating room: number of planned surgeries, planned duration metrics (minutes), and procedure diversity

PlannedOR	Count	Percentage	Mean	Median	SD	CV	IQR	P95	UniqueProcedures
GO11	7460	9.4%	77.1	67	51.9	0.67	50	154	610
GO14	6889	8.7%	51.6	40	50.7	0.98	47	120	228
GO01	6587	8.3%	61.7	36	82.4	1.33	30	178.7	195
GO17	5496	6.9%	75.6	78	48.6	0.64	60	147	274
GO18	5297	6.7%	74.9	72	53.5	0.71	49	150	280
GO06	5228	6.6%	86.4	48	90.8	1.05	52	248.9	164
GO05	4876	6.1%	103.9	73	90.1	0.87	68	245	297
GO07	4664	5.9%	78	69	60.7	0.78	40	160	311
GO04	4508	5.7%	104.7	72	92.5	0.88	63	289	263
GO15	4313	5.4%	89.1	80	54.2	0.61	69	164	323
GO03	4095	5.2%	99.9	83	69.1	0.69	57	200	184
GO16	4094	5.2%	95.3	101	45.4	0.48	50	150	283
GO02	3500	4.4%	109.3	80	93.8	0.86	83	279	240
GO13	3285	4.1%	138.9	110	94.8	0.68	74	305	188
GO12	2904	3.7%	144.9	110	95.4	0.66	91	320.8	159
GO09	2643	3.3%	144.5	124	102.7	0.71	123	322	263
GO08	1765	2.2%	107	76	81.7	0.76	101	270	151
GO10	1748	2.2%	286.9	318	146.1	0.51	178.5	480	177

The distribution of planned surgeries across operating rooms is highly uneven. A small number of rooms show clearly higher planned activity than the others, as reflected by the tallest bars on the left side of the figure. After these high volume rooms, the number of planned surgeries decreases gradually across the remaining operating rooms. Toward the right side of the chart, several rooms display much lower planned activity, with visibly short bars. This decline illustrates the concentration of planned surgical volume in a limited subset of operating rooms, while many others are used less intensively. It is notable that GO11 shows the highest planned activity. Figure 6 shows the full distribution across rooms.

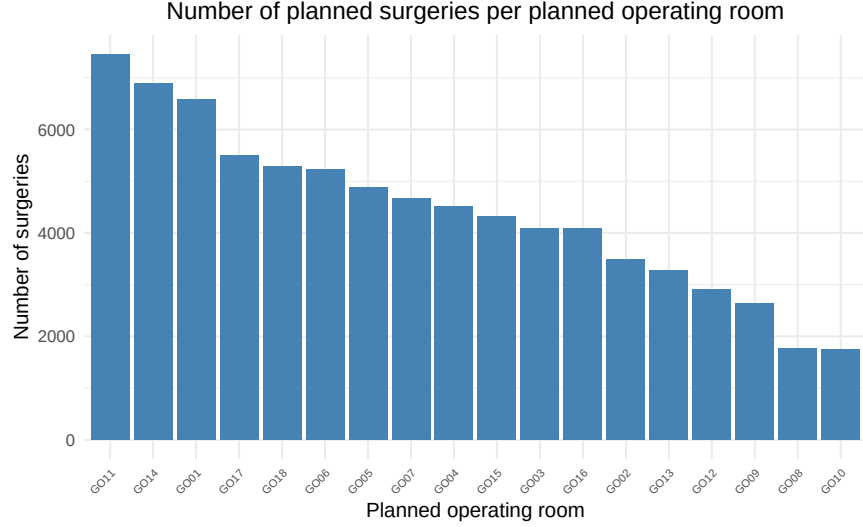


Figure 6: Number of planned surgeries per planned operating room

Actual surgery information

The actual surgery information provides an overview of how surgical activity was ultimately carried out in practice. By examining actual durations, operating room use, and volume distribution, this section highlights how the execution of surgeries compares to the initial plan. Understanding these real-world patterns is essential for interpreting deviations, overtime, and workflow efficiency in the following analyses.

ActualOR

The distribution of mean durations is broad, with some rooms showing relatively short average procedure times and others much longer ones, as visible from the variation in the mean values in Table 8. This indicates substantial differences in typical case lengths.

The coefficient of variation, shown in Table 8, displays considerable variability between rooms. The diversity of procedure types also differs strongly, with some rooms registering only a few unique procedures and others several hundred. This large variation shows that some operating rooms handle highly specialized activities, while others are used for a wide mix of procedures. Overall, these results demonstrate that actual OR utilization, like the planning data, is unevenly distributed across rooms, with notable heterogeneity in both volume and case complexity.

Table 8: Per operating room: number of actual surgeries, duration metrics (minutes), and procedure diversity

ActualOR	Count	Percent	Mean	Median	SD	CV	IQR	P95	UniqueProc
GO11	7482	9.4%	83.5	72	53.2	0.64	54	180	611
GO14	6885	8.7%	50.6	41	42.8	0.85	44	120	226
GO01	6577	8.3%	59.6	38	70.1	1.18	31	165	197
GO17	5480	6.9%	70.2	67	44.6	0.64	47	150	278
GO18	5293	6.7%	70.5	63	43.5	0.62	48	151	281
GO06	5217	6.6%	82.4	47	82.2	1	60	266	166
GO05	4886	6.2%	104	77	78.6	0.76	77	247	304
GO07	4658	5.9%	77.1	66	52	0.67	46	171.1	314

ActualOR	Count	Percent	Mean	Median	SD	CV	IQR	P95	UniqueProc
GO04	4518	5.7%	102.7	72	87.2	0.85	68	277.1	266
GO15	4323	5.4%	86.5	76	50.8	0.59	66	178.9	324
GO16	4098	5.2%	92.6	96	40.8	0.44	53	155	282
GO03	4094	5.2%	96.7	84	55.2	0.57	56	196	185
GO02	3502	4.4%	110.2	86	92.2	0.84	87	281.9	240
GO13	3298	4.2%	138.5	112	88.2	0.64	85	311.1	193
GO12	2884	3.6%	145.7	113	95	0.65	91.2	341	159
GO09	2637	3.3%	143.6	119	96.9	0.67	128	337	262
GO08	1777	2.2%	111.6	80	86.2	0.77	100	291.2	158
GO10	1743	2.2%	284.9	302	145.4	0.51	187	498.8	176

The distribution of actual surgeries across operating rooms closely mirrors the planning data. A small number of rooms show the highest levels of activity, as indicated by the tallest bars on the left side of the figure. The remaining rooms display a gradual decline in volume, with most rooms having visibly lower numbers of surgeries as the sequence progresses. Toward the end of the chart, several rooms record only limited activity, shown by the shortest bars. This uneven distribution suggests that operating room usage is concentrated in a limited subset of rooms, while others are used less intensively or serve more specialized purposes. Figure 7 shows the full distribution.

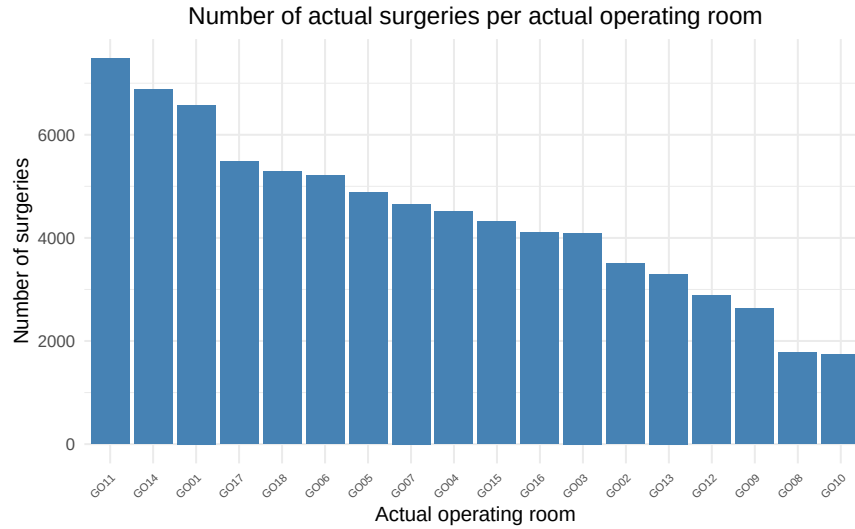


Figure 7: Number of actual surgeries per actual operating room

A closer look on GO10 A more detailed analysis of GO10 shows that, although this operating room is intended for a specific subset of procedures, the number of unique procedures is unexpectedly high. Only a small number of procedure types occur frequently, with six procedures performed more than fifty times. In contrast, the majority of procedures are observed only a limited number of times, with many appearing between two and five times, and a large group occurring only once.

This distribution indicates that, despite the specialized nature of GO10, there is still considerable variation in the types of procedures performed. The large number of unique procedures suggests that the room is not exclusively dedicated to a narrowly defined set of standardized interventions, but rather accommodates a broader range of cases than initially expected.

Table 9: Frequency distribution of procedure types in GO10, with number of unique procedures and three example procedures per frequency group

FrequencyGroup	NumberOfUniqueProcedures	ThreeExampleProcedures
>50	6	COR.AORTA BYPASS GRAFT V., AORTAKLEP VIA MINI-STERNOTOMIE, PORT ACCES MITRALISKLEP
21–50	6	RAL MIDCAB, THORACOSCOPISCHE ABLATIE MINI MAZE, MVR
11–20	6	ANDERE, AVR_TAX, WONDDEBRIDEMENT
6–10	15	PM/DEFIB/LEAD, EXTR. CORP.MEMBRAAN OXYG., EXTRACTIES
2–5	54	BIOPSIE KLIER, EXC MALIGNIE HUIDLETSEL MT RECONST HUIDGR LICHAAM, PORT-A-CATH
1	89	ABDOMINAAL ANEURYSMA, AMPUTATIE BOVENBEEN, ANALE EXPLORATIE

DurationMinutes

The average observed surgery duration is 92.9 minutes, with a median of 71 minutes, showing that most procedures are relatively short but a smaller group of long surgeries increases the mean. The interquartile range of 69 minutes indicates substantial variability across cases. Most surgeries fall well below the upper extremes of the distribution, while a small number extend to much longer durations, with values up to 1043 minutes. At the upper end of the distribution, the ninety fifth percentile equals 253 minutes, the ninety ninth percentile 411 minutes, and the ninety nine point nine percentile 630 minutes. This confirms a strong right skewed distribution, where the majority of surgeries are short, and long duration cases represent rare but significant outliers. Table 10 summarises these statistics.

Table 10: Actual surgery duration (minutes), summary statistics

Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
92.9	71	79.6	69	1	1043	253	411	630

The distribution of actual surgery durations follows a clear right skewed pattern. Most procedures appear in the shorter duration ranges, where the bars are visibly highest, and the frequency decreases steadily as durations increase. Only a small fraction of cases appears in the longest duration ranges. This concentration of short procedures reflects the overall composition of surgical activity, where brief interventions dominate and long operations are relatively rare. The trimmed view at the ninety ninth percentile, shown in Figure 8, highlights the main body of the data without distortion from extreme outliers.

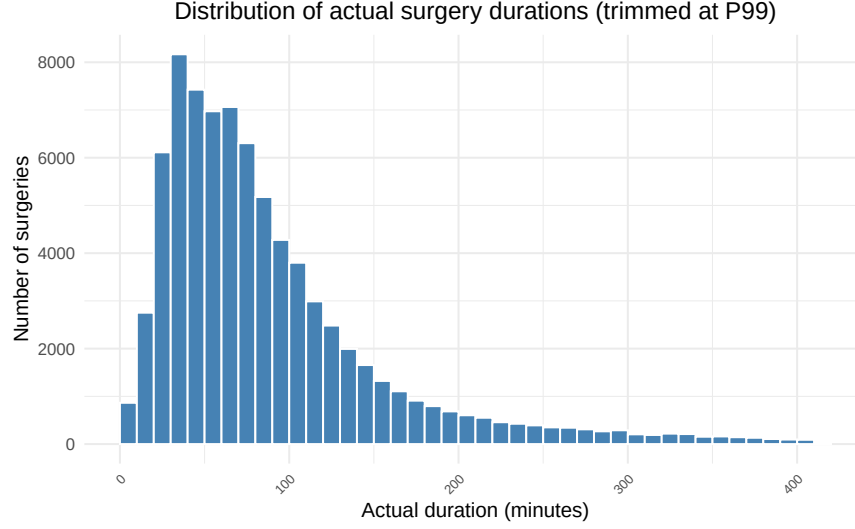


Figure 8: Distribution of actual surgery durations (trimmed at P99)

Deviation metrics

Deviation metrics capture the numerical differences between planned and actual performance at different points in the workflow. They quantify how accurately start times, end times, and total durations match the original plan. Analysing these deviations helps identify where delays or overruns of a particular surgery most often occur and whether they are systematic or random. This section provides the foundation for understanding the root causes of schedule inaccuracy.

StartDiff

The difference between actual and planned start times shows substantial variation across the dataset. When distinguishing between urgency types, clear differences emerge. Elective surgeries start on average 25.1 minutes later than planned, with a median delay of 9 minutes, whereas non elective surgeries show a much larger average delay of 125.7 minutes and a median of 27 minutes. The very high standard deviation reflects the presence of extreme values, which likely represent registration anomalies rather than genuine scheduling deviations. Table 11 summarises these deviation metrics.

For elective cases, 28.8 percent start more than 30 minutes late, while 7.9 percent begin more than 30 minutes early. For non elective cases, 47.5 percent start more than 30 minutes late and 10.0 percent start more than 30 minutes early. This indicates that a noticeable share of surgeries deviates from the planned start time, with delays being more common than early starts. These figures provide a first quantitative indication of how frequently the operating schedule diverges from the plan at the start of the day.

Table 11: Difference between actual and planned start time in minutes by urgency type

UrgencyType	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	> 30 min late	< -30 min early
electief	25.1	9	106.2	40	-1013	1769	114	572	1093	28.8%	7.9%
niet-electief	125.7	27	275.9	96	-1366	1863	779.2	1134	1386	47.5%	10.0%

The distribution of start time deviations is centred around zero but shows a clear skew toward positive values, indicating that delays are more frequent than early starts. The tallest bar near zero reflects that many surgeries begin close to the planned time, while the bars extend outward on both sides, forming long tails that illustrate the presence of substantial deviations. The right tail is visibly more pronounced than the left, confirming that larger delays occur more often than equally large early starts. Overall, the pattern suggests that day to day planning captures the general timing well, although start time punctuality remains difficult to maintain for a considerable number of surgeries. Figure 9 shows the distribution grouped in ten minute intervals.

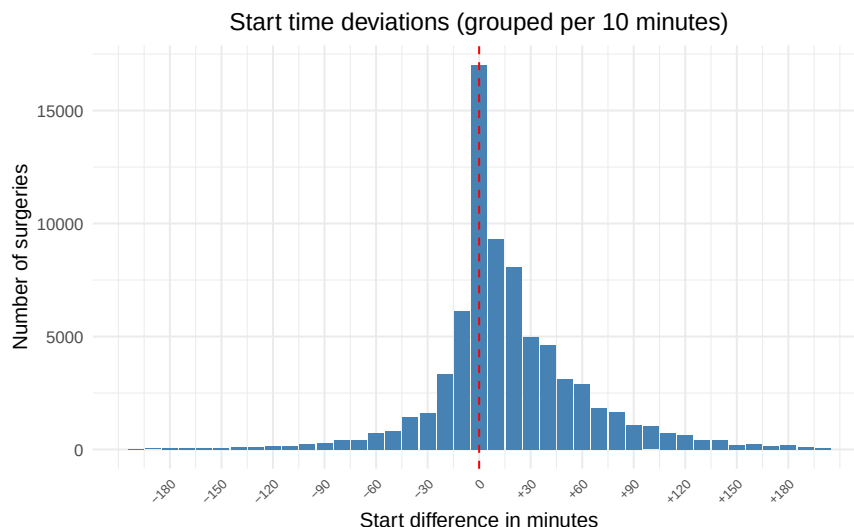


Figure 9: Start time deviations grouped per ten minutes

Among the one percent most extreme start time deviations, a small group of procedure types appears more frequently than others. Table 12 shows that POLS FRAKTUUR DISTAAL RADIUS and HEUP FRACTUUR TFNa are the most frequent, with 64 and 50 cases respectively. Other procedures, including SECTIO, ENKEL FRAKTUUR AO KLEIN FRAGMENT, HEUP / TOTALE HEUPPROTHESE, PEES HAND (VINGER) VOET HECHTING, LAP.APPENDIX, VINGER FRAKTUUR PINNING, VOORARM FRAKTUUR PINNING, and POLS FRAKTUUR REDUCTIE GESLOTEN, also appear regularly among the outliers. This indicates that extreme deviations occur across a diverse range of surgical activities, with frequency largely reflecting overall procedure volume.

Table 12: Top 10 procedure types among the one percent most extreme start time deviations

ProcedureName	n
POLS FRAKTUUR DISTAAL RADIUS	64
HEUP FRACTUUR TFNa	50
SECTIO	26
ENKEL FRAKTUUR AO KLEIN FRAGMENT	25
HEUP / TOTALE HEUPPROTHESE	22
PEES HAND (VINGER) VOET HECHTING	19
LAP.APPENDIX	16
VINGER FRACTUUR PINNING	16
VOORARM FRAKTUUR PINNING	15

ProcedureName	n
POLS FRACTUUR REDUCTIE GESLOTEN	13

The analysis of operating rooms with the most extreme start time deviations shows that a small number of rooms account for a large share of the outliers. GO11 stands out clearly, with 713 extreme cases, far exceeding any other room. Other frequently affected rooms include GO04, GO12, GO07, and GO05, each with single digit counts. The remaining rooms show smaller counts. This pattern suggests that start time deviations are not evenly distributed across the operating theatre complex but tend to cluster in a limited number of rooms. Table 13 provides an overview of the ten operating rooms most frequently involved.

Table 13: Top 10 operating rooms among the one percent most extreme start time deviations

ActualOR	n
GO11	713
GO04	10
GO12	9
GO07	8
GO05	7
GO06	7
GO09	7
GO13	6
GO17	6
GO03	5

Among the most extreme one percent of start time deviations, the vast majority are late starts. A total of 780 cases, or 98.1 percent, begin later than planned, while only 15 cases, or 1.9 percent, start earlier. This clear imbalance confirms that extreme deviations almost always occur in the direction of delay rather than early execution. It suggests that structural or operational factors are more likely to cause postponements than premature starts, making delayed openings the dominant source of major scheduling disruptions. Table 14 summarises this distribution.

Table 14: Distribution of direction, early versus late, among the one percent most extreme cases

Direction	n	Percentage
Early	15	1.9%
Late	780	98.1%

EndDiff

The difference between actual and planned end times shows a clear tendency toward delayed finishes. On average, surgeries end 38.7 minutes later than planned, with a median delay of 14 minutes. The large standard deviation of 153.5 minutes reflects the presence of extreme timestamps that fall outside realistic operating times. In total, 36.0 percent of cases finish more than 30 minutes late, while 13.4 percent end more than 30 minutes early. These results indicate that moderate delays are common, whereas very large deviations occur in a more limited number of cases. Table 15 summarises these results.

Table 15: Difference between actual and planned end time in minutes

Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	> 30 min late	< -30 min early
38.7	14	153.5	60	-1804	1863	182	783	1253	36.0%	13.4%

The distribution of end time deviations is centred around zero but slightly skewed toward positive values, indicating that delays are more frequent than early completions. The tallest bars appear near zero, showing that many surgeries finish close to the planned time, while the distribution tapers off gradually in both directions. The right tail is visibly more extended than the left, confirming that some procedures run substantially longer than expected. This pattern closely resembles the distribution observed for start time deviations, suggesting a similar overall pattern in scheduling deviations. Figure 10 shows the distribution grouped in ten minute intervals.

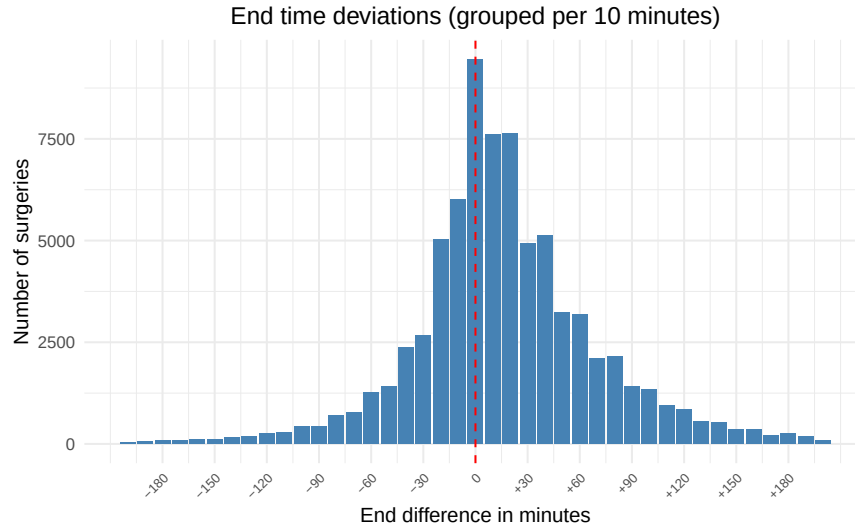


Figure 10: End time deviations grouped per ten minutes

Among the one percent most extreme end time deviations, a small number of procedure types appear more frequently than others. Table 16 shows that POLS FRAKTUUR DISTAAL RADIUS and HEUP FRACTUUR TFNa are the most frequent, with 60 and 48 cases respectively. Other procedures, including SECTIO, ENKEL FRAKTUUR AO KLEIN FRAGMENT, HEUP / TOTALE HEUPPROTHESE, LAP.APPENDIX, PEES HAND (VINGER) VOET HECHTING, VINGER FRAKTUUR PINNING, VOORARM FRAKTUUR PINNING, and EXPLORATIEVE LAPARASCOPIE, also appear regularly among the outliers. This variety indicates that large deviations occur across different kinds of interventions rather than being specific to a single surgical category.

Table 16: Top 10 procedure types among the one percent most extreme end time deviations

ProcedureName	n
POLS FRAKTUUR DISTAAL RADIUS	60
HEUP FRACTUUR TFNa	48
SECTIO	26

ProcedureName	n
ENKEL FRAKTUUR AO KLEIN FRAGMENT	24
HEUP / TOTALE HEUPPROTHESE	24
PEES HAND (VINGER) VOET HECHTING	19
VINGER FRACTUUR PINNING	16
VOORARM FRAKTUUR PINNING	16
EXPLORATIEVE LAPARASCOPIE	12
LAP.APPENDIX	12

A small number of operating rooms account for most extreme end time deviations. GO11 clearly dominates, with 678 cases, while the other rooms in the top ten show much lower counts. These include GO05, GO04, GO06, and GO12, each with values of around 10 to 12 cases, and several rooms with even fewer occurrences. This suggests that end time overruns are not evenly distributed but tend to cluster in a limited set of operating rooms. Table 17 provides an overview of the ten rooms most frequently involved.

Table 17: Top 10 operating rooms among the one percent most extreme end time deviations

ActualOR	n
GO11	678
GO04	12
GO05	12
GO06	12
GO03	10
GO12	10
GO09	9
GO07	8
GO13	8
GO14	7

Among the most extreme end time deviations, the vast majority of cases end later than planned. A total of 724 cases, or 91.2 percent, finish late, compared to 70 cases, or 8.8 percent, that end earlier than planned. This consistent asymmetry confirms that schedule overruns are the dominant pattern, similar to what was observed for start times. Large early completions remain rare, suggesting that delays accumulate more easily than time savings during the surgical process. Table 18 summarises this distribution.

Table 18: Distribution of direction, early versus late, among the one percent most extreme end time deviations ### DurationDiff

Direction	n	Percentage
Early	70	8.8%
Late	724	91.2%

When distinguishing between longer and shorter cases, a clear asymmetry appears. A total of 45.7 percent of surgeries take longer than planned, with an average overrun of 22.6 minutes and a median of 14 minutes. In contrast, 54.3 percent finish sooner, with an average gain of 21.2 minutes and a median of 12 minutes. The standard deviation is considerably larger for cases that finish sooner (52.4 minutes) than for those that take longer (27.5 minutes), indicating that some procedures end far earlier than planned, while overruns

are more tightly concentrated. Overall, most operations remain reasonably close to their planned duration, suggesting that duration estimates are generally reliable. Table 19 summarises these patterns.

Table 19: Difference between actual and planned duration, absolute minutes, separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	Percent
Longer than planned	22.6	14	27.5	23	1	651	73	133	245.7	45.7%
Shorter than planned	21.2	12	52.4	19	0	1374	58	122	856	54.3%

Among the one percent most extreme duration deviations, a small number of procedures appear more frequently than others. Table 20 shows that COR.AORTA BYPASS GRAFT V. is the most common, with 60 extreme cases in total, consisting of 35 longer and 25 shorter procedures. Other procedures that recur among the outliers include AVR, P.L.I.F. MET BRAINLAB AERO, EXPLORATIEVE LAPARASCOPIE, EXPLORATIEVE LAPAROTOMIE, CABG OFF PUMP, THORACOSCOPISCHE LOBECTOMIE, BENTALL, DEBULKING MET HIPEC, and TREPANATIE TUMOR. These findings show that extreme deviations occur across a diverse set of interventions rather than being limited to a single surgical category.

Table 20: Top 10 procedure types with the highest number of extreme duration deviations

ProcedureName	Longer	Shorter	Total
COR.AORTA BYPASS GRAFT V.	35	25	60
AVR	8	14	22
P.L.I.F. MET BRAINLAB AERO	13	6	19
EXPLORATIEVE LAPARASCOPIE	14	3	17
EXPLORATIEVE LAPAROTOMIE	12	1	13
CABG OFF PUMP	2	11	13
BENTALL	5	8	13
THORACOSCOPISCHE LOBECTOMIE	9	3	12
DEBULKING MET HIPEC	6	6	12
TREPANATIE TUMOR	6	6	12

A limited number of operating rooms account for the majority of extreme duration deviations. GO10 stands out with 146 extreme cases, followed by GO11 with 82 cases. Other frequently affected rooms include GO09, GO05, and GO12, as shown by their totals in Table 21. The remaining rooms in the top ten register lower but still notable counts of extreme deviations. In some rooms, longer than planned procedures slightly outnumber shorter ones, while in others the opposite pattern appears. Overall, the largest mismatches between planned and actual duration are concentrated in a small set of operating rooms. Table 21 summarises these operating rooms.

Table 21: Top 10 operating rooms with the highest number of extreme duration deviations

ActualOR	Longer	Shorter	Total
GO10	75	71	146
GO11	57	25	82
GO05	35	37	72

ActualOR	Longer	Shorter	Total
GO09	37	33	70
GO12	33	25	58
GO08	34	13	47
GO04	17	29	46
GO13	21	23	44
GO01	17	24	41
GO06	15	25	40

Among the most extreme one percent of duration deviations, 405 cases, or 50.4 percent, are longer than planned, while 398 cases, or 49.6 percent, are shorter. This distribution shows a relatively small asymmetry, indicating that extreme deviations occur in both directions. Table 22 summarises this distribution.

Table 22: Distribution of direction among the one percent most extreme duration deviations

Direction	n	Percentage
Longer	405	50.4%
Shorter	398	49.6%

percentage_deviation

When expressing the differences between planned and actual durations in relative terms, a clear asymmetry emerges between longer and shorter cases. Surgeries that last longer than planned deviate on average by 32.8 percent, with a median deviation of 21.1 percent, while procedures that finish sooner deviate by 20 percent on average and 17 percent at the median. This indicates that overruns tend to be larger in magnitude than underruns, even though both occur frequently. The interquartile ranges of 32 percent for longer cases and 20.6 percent for shorter cases show substantial variability across procedures. While many surgeries remain within moderate limits, the upper percentiles reveal that some operations deviate substantially from expectations, with the ninety fifth percentiles equal to 100 percent for longer cases and 49.5 percent for shorter cases. Table 23 summarises these patterns.

Table 23: Relative deviation, separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
Shorter than planned	20	17	15.6	20.6	0	98.5	49.5	68.2	95.4
Longer than planned	32.8	21.1	40.7	32	0.2	1050	100	187.2	400

The distribution of percentage deviations is centred around zero, indicating that many surgeries finish close to their planned duration. The peak is slightly shifted toward negative values, showing that more procedures finish earlier than planned. The distribution tapers gradually on both sides, with visibly longer tails for positive deviations. The narrow central peak suggests that planning is generally accurate, although the wider spread of the bars shows that notable deviations still occur in a substantial number of cases. Figure 11 displays the distribution grouped per ten percent.

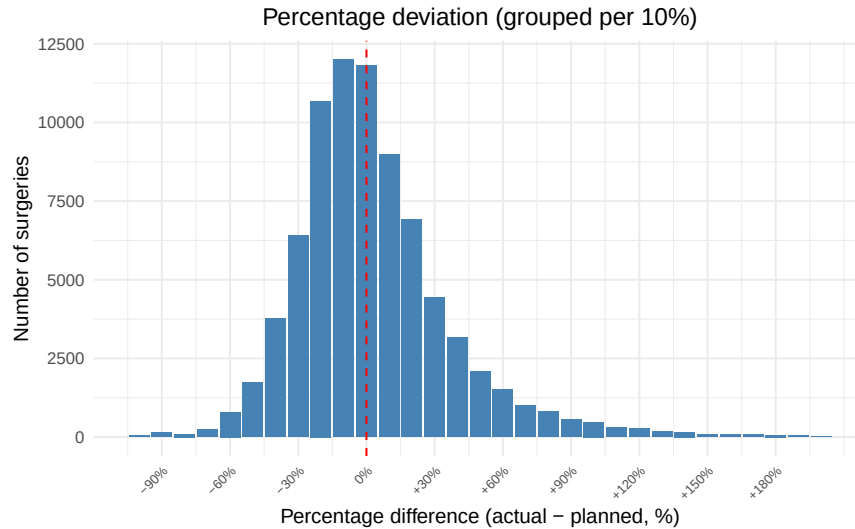


Figure 11: Percentage deviation (grouped per 10 percent)

The procedures with the highest average relative deviations show considerable differences between planned and actual durations. As shown in Figure 12, the procedure with the largest mean deviation is I & D VAN ADEONFLEGMOENE, followed by ELLEBOOG FRACTUUR REDUCTIE and several other procedures that also display noticeably high deviations. These results indicate that for some procedure types the planned duration does not always reflect the time actually required in practice. Further analysis would be needed to determine whether this relates to scheduling conventions or other operational factors. Figure 12 shows the top twenty procedure types with the highest mean relative deviation.

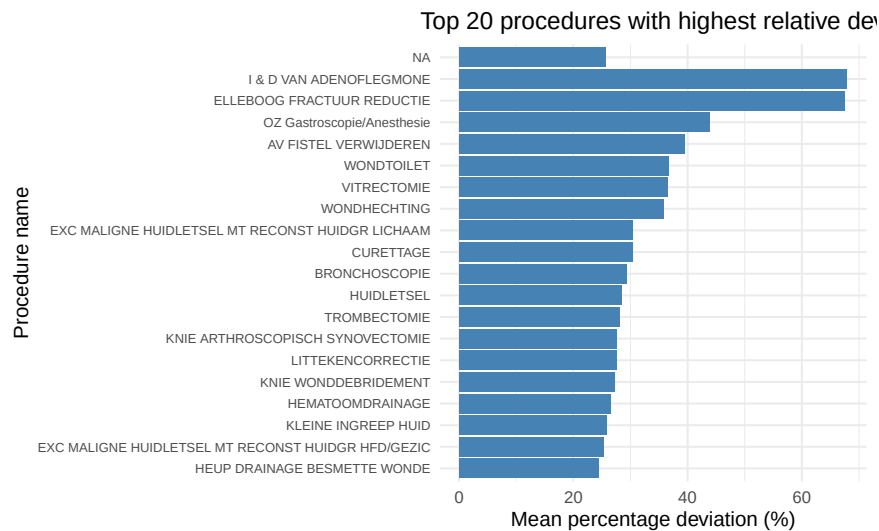


Figure 12: Top 20 procedure types with highest relative deviation

The average relative deviation varies noticeably by weekday. From Monday to Friday, the bars remain low and fairly consistent, indicating stable planning accuracy during regular operating days. In contrast, Saturday and Sunday show markedly higher deviations, with both weekend days displaying substantially taller bars than any weekday. This pattern suggests that weekend cases are less predictable in duration. Figure 13 shows the mean relative deviation for each weekday.

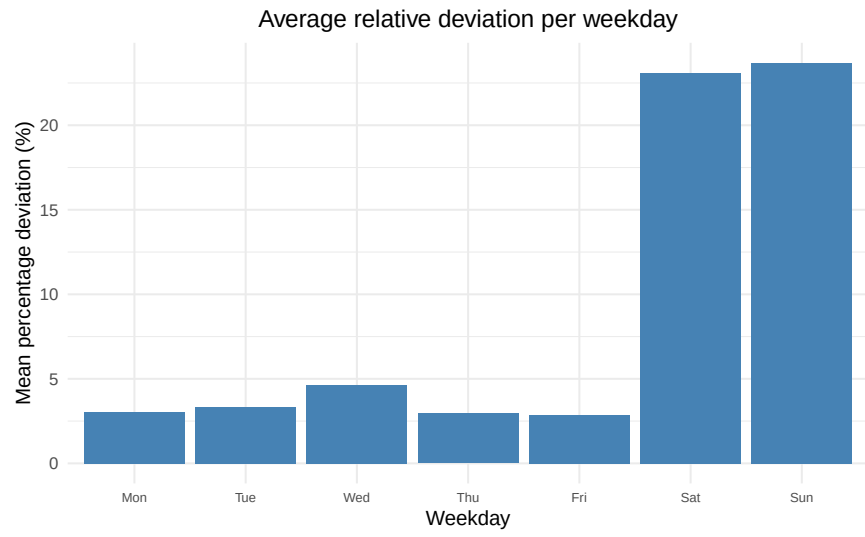


Figure 13: Average relative deviation per weekday

Conclusion

The exploratory analysis provides an overview of surgical activity at Ziekenhuis Oost Limburg (Genk) across multiple dimensions of time, procedure type, and operating room planning. The results show that most surgeries are elective and performed during regular daytime hours, with short hospital stays and a strong presence of both inpatient and day surgery. This reflects a high throughput surgical system mainly oriented toward planned interventions.

The descriptive statistics also show considerable variability in planned and observed metrics between operating rooms and procedure types. The wide range of planned durations, the uneven distribution of workload across rooms, and the long tail of hospital stays highlight the operational complexity behind the overall averages. These patterns underline the need for data driven performance metrics that go beyond simple utilization or volume counts.

The insights from this baseline analysis form the foundation for more advanced performance evaluation. By quantifying deviations between planned and observed activity, it becomes possible to identify where scheduling accuracy and resource use can be improved.